

FULL STACK WEB DEVELOPMENT USING MERN			
Course Code	25MCA301	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the MVC architectural pattern and its role in web development. Articulate what React is and why it is useful.		
2.	Utilize modern JavaScript features like ES6+ classes, modules, and arrow functions. Making use of controlled components, create and verify forms. Use a JavaScript package manager (either npm or Yarn)		
3.	Build reusable React components and implement state management effectively. Making HTTP requests to read or modify data.		
4.	Develop and integrate RESTful APIs using Express.js and MongoDB. Implement client-side routing using React Router.		
5.	Students able to create CRUD operations, routing mechanisms, and UI components.		
UNIT-I			
Introduction to the MERN stack:			04 Hours
Introduction, The MVC architectural Pattern. MERN Components React, Node.js, Express, MongoDB, Advantages and real-world applications of MERN Stack, understanding Isomorphic and types			
MERN Components			04 Hours
Advantages and real-world applications of MERN Stack. Understanding Isomorphic and types. Welcome to React- Obstacles and Roadblocks, Reacts Future			
UNIT-II			
Understanding React			04 Hours
Server setup, setting up a Development Environment -NVM, Node Js, Project, NPM, Express, Understanding JSX and it's compilation process, Separate Script File, Transform, Automate, React Library			
Web Server:			04 Hours
React Components-React classes, Composing components, passing data- using properties, property validation, using children's Dynamic, React Lifecycle Methods.			
UNIT-III			
Understanding React State:			04 Hours
React State – Setting state, Event handling, communicating from child to parent, Stateless components, Designing Components-state vs props			
React Components			02 Hours
component hierarchy communication, Introduction to React Context API.			
UNIT-IV			
Building RESTful APIs			05 Hours
REST , HTTP method as Actions, JSON, Express, Request Object, Response objects, Middleware, The list API , The create API, Using the LIST API, Using the Create API, Error Handling.			

Mongo DB:										05 Hours		
Mongo DB: Basics, Schema initialization , MongoDB Node.JS Driver, callbacks, promises, Reading from MongoDB, Writing to MongoDB												
UNIT-V												
Working with React Router and forms:										04 Hours		
Routing Techniques, Simple Routing, Route parameters, Route Query String, Programmatic Routing, Nested Routes, Browser history.												
Forms:										04 Hours		
Forms, Filter form, Get API, Edit page, Building UI Components and Forms, update API, Delete API.												
Course Outcomes: At the end of this course, the students will be able to:												
1.	Understand the concepts of MVC components and architecture and Implement node js, Express and MongoDB.											
2.	Use JSX transformation and automatic recompiling for better React development.											
3.	Develop React components and manage UI state effectively.											
4.	Implement State management and event handling in React applications.											
5.	Build Restful APIs, perform CRUD operations, and integrate UI components efficiently.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes												
Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes											1	2
25MCA301.1			3	3	2	2	2	1	1	1	3	2
25MCA301.2			2	3	3	3	3	1	1	1	3	3
25MCA301.3			2	3	3	3	3	1	1	1	3	3
25MCA301.4			2	3	3	3	3	1	1	1	3	3
25MCA301.5			3	3	3	3	3	2	1	1	3	3
1: Low 2: Medium 3: High												
TEXTBOOKS:												
1.	Pro MERN Stack, Vasan Subramanian, 2019, ISBN-13(pbk):978-1-4842-2653-7											
2.	MERN Quick Start Guide, Eddy Wilson Iriarte Koroloiva, 2018, PACKT Publication, ISBN 978-1-78728-108-0											
REFERENCE BOOKS:												
1.	Full stack React, Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, and Tyler McGinnis, 2017.											

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING			
Course Code	25MCA302	Course Type	PCC
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Mathematical Foundation for computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand the advantages and types of Artificial Intelligence		
2.	To understand machine learning, the need for machine learning, its types and data preprocessing		
3.	To understand Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning and Evaluating Regression model performance.		
4.	To understand Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms.		
5.	To learn deep learning concepts, Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks.		
UNIT - I			
Artificial Intelligence:			06 Hours
What is AI; Why Artificial Intelligence; Goals of AI; Advantages and disadvantages of AI; Applications of AI, Types of AI, Strategies for State Space Search (Representation of State Space as Graphs), Introduction to Heuristics			
UNIT - II			
Machine learning:			08 Hours
What is ML; Need for ML; Challenges and applications of ML; classification: Supervised, Unsupervised, Reinforcement learning, AI vs Machine learning, Data Preprocessing.			
UNIT - III			
Regression:			10 Hours
Simple, Linear Regression, Multiple Regression, Polynomial Regression, Logistic Regression, Decision Trees, Support Vector Machines, Naive Bayesian Learning, Evaluating Regression model performance.			
Unit - IV			
Clustering:			08 Hours
K-Means Clustering, Hierarchical Clustering, K-NN, Principal Component Analysis, Reinforcement Learning			
Unit - V			
Deep Learning and Artificial Neural Networks:			08 Hours
Introduction to deep learning, The Neuron, How do Neural network learn and work? Gradient descent, Back Propagation, ANN in python, Convolutional Neural Networks.			
Course Outcomes: At the end of the course student will be able to			
1.	Outline the fundamental concepts of Artificial Intelligence and its types.		
2.	Analyze the need for machine learning and data preprocessing		
3.	Apply Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning for data analysis and prediction.		

4.	Apply machine learning techniques like Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms for data analysis and classification.
5.	Apply deep learning concepts like Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks for classification.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA302.1	3	3	3	3	3	-	3	-	3	3
25MCA302.2	3	3	3	3	3	-	3	-	3	3
25MCA302.3	3	3	3	3	3	-	3	-	3	3
25MCA302.4	3	3	3	3	3	-	3	-	3	3
25MCA302.5	3	3	3	3	3	-	3	-	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, "Machine Learning", 1st Edition, McGraw Hill Education, 2017.
3.	Subramanya Bhat, Machine Learning – A Simplified Approach, 1 st Edition, Narosa Publishing House Pvt. Ltd., 2025
4.	Elaine Rich, Kevin K and S B Nair, "Artificial Intelligence", 3 rd Edition, McGraw Hill Education, 2017

REFERENCE BOOKS:

1.	Saroj Kaushik, Artificial Intelligence, Cengage learning, 2014
2.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
3.	Data Science: Concepts and Practice - By Vijay Kotu, Bala Deshpande 2nd edition, Publisher -Morgan Kaufmann, 2018
4.	Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2 nd Edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016
5.	Data Science for Dummies, Lillian Pierson, John Wiley & Sons, 21-Feb-2017

CLOUD COMPUTING AND BIG DATA ANALYTICS			
Course Code:	25MCA303	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Prerequisites	Student must have some basic knowledge of Core Java, database concepts, and any of the Linux operating system flavors		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the types, characteristics and challenges with digital data		
2.	Understand the concept of Big data Analytics and technology landscape		
3.	Understand the use of open source software framework called Hadoop		
4.	Understand different cloud computing service models and cloud security		
5.	Understand various technologies of cloud computing.		
UNIT - I			
Introduction to Big Data:			10 Hours
Classification of Digital Data, Structured Data, Sources of Structured Data, Ease of Working with Structured Data, Semi-Structured Data, Sources of Semi-Structured Data, Unstructured Data, Issues with “Unstructured” Data, How to Deal with Unstructured Data. Characteristics of Data, Evolution of Big Data, Definition of Big Data, Volume, Velocity, Variety, Challenges of Big Data, Other Characteristics of Data Which are Not Definitional Traits of Big Data, Traditional Business Intelligence (BI) versus Big Data, A Typical Data Warehouse Environment, A Typical Hadoop Environment, Coexistence of Big Data and Data Warehouse. Exploring the Big Data Stack			
Unit - II			
Introduction to Big Data Analytics :			10 Hours
Big Data Analytics, Classification of Analytics, Greatest Challenges that Prevent Businesses from Capitalizing on Big Data, Top Challenges Facing Big Data, Importance of Big Data Analytics, Terminologies Used in Big Data Environment, In Memory Analytics, In Database Processing, Symmetric Multiprocessor System, Massively Parallel Processing, Difference between Parallel versus Distributed Systems, Shared Nothing Architecture, Consistency, Availability, Partition Tolerance (CAP), Basically Available Soft State Eventual Consistency (BASE) Basic analytics, Advanced analytics, Operationalized analytics, Monetizing analytics. Modifying Business Intelligence Products to Handle Big Data, Studying Big Data Analytics			
Unit - III			
Hadoop Techniques:			10 Hours
Introducing Hadoop, RDBMS versus Hadoop, Hadoop versus SQL, Distributed Computing Challenges, Key Aspects of Hadoop, Hadoop Components. Features of Hadoop, Key Advantages of Hadoop, Versions of Hadoop, Hadoop 1.0, Hadoop 2.0, Overview of Hadoop Ecosystems, High Level Architecture of Hadoop, Hadoop Distributed File System, HDFS Daemons, Anatomy of File Read, Anatomy of File Write, Processing Data with Hadoop, MapReduce Daemons, MapReduce Example, Managing Resources and Application with Hadoop YARN, Limitations of Hadoop 1.0 Architecture, HDFS Limitation, Hadoop 2: HDFS, Hadoop 2 YARN: Taking Hadoop Beyond Batch, Hadoop Ecosystem. Introduction to NoSQL, NewSQL Uses, Features and Types, Disadvantages and Application of NoSQL.			

Unit - IV																																																																																								
Cloud Computing definition, Cloud service models (IaaS, PaaS & SaaS) :										10 Hours																																																																														
Cloud deployment models (Public, Private, Hybrid and Community Cloud), Private & Public Cloud Definition, Characteristics of Private Cloud, Private Cloud deployment models, Private Cloud Building blocks namely Physical Layer, Virtualization Layer, Cloud Management Layer, When to opt for Public Cloud, Public Cloud Service Models.																																																																																								
Virtualization																																																																																								
Basics of virtualization, Virtualization technologies, Server virtualization, VM migration techniques, Role of virtualization in Cloud Computing.																																																																																								
Unit - V																																																																																								
Cloud Tools and Applications:										10 Hours																																																																														
Aneka PaaS; Open stack: Introduction to open stack; Components of open stack; Amazon web services; Google AppEngine; Microsoft Azure; Scientific applications: Healthcare; Biology; Geo-Science, Business and Consumer applications: ARM & ERP; Productivity; Social networking.																																																																																								
Course Outcomes: At the end of the course student will be able to																																																																																								
1.	Classify the digital data																																																																																							
2.	Understand the top challenges facing big database																																																																																							
3.	Study components and advantages of Hadoop																																																																																							
4.	Understand Cloud deployment models																																																																																							
5.	Identify enabling technologies of cloud computing.																																																																																							
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes																																																																																								
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>PSO</td><td>PSO</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>2</td></tr><tr><td>25MCA303.1</td><td>-</td><td>3</td><td>-</td><td>-</td><td>3</td><td>-</td><td>3</td><td>3</td><td>1</td><td>1</td></tr><tr><td>25MCA303.2</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td><td>2</td><td>3</td><td>3</td><td>2</td><td>1</td></tr><tr><td>25MCA303.3</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td></tr><tr><td>25MCA303.4</td><td>-</td><td>3</td><td>-</td><td>3</td><td>3</td><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td></tr><tr><td>25MCA303.5</td><td>-</td><td>3</td><td>2</td><td>2</td><td>2</td><td>3</td><td>3</td><td>3</td><td>1</td><td>2</td></tr></table>												Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	↓ Course Outcomes									1	2	25MCA303.1	-	3	-	-	3	-	3	3	1	1	25MCA303.2	-	3	3	3	3	2	3	3	2	1	25MCA303.3	-	3	3	3	3	-	3	3	3	3	25MCA303.4	-	3	-	3	3	2	3	3	2	3	25MCA303.5	-	3	2	2	2	3	3	3	1	2
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25MCA303.4	-	3	-	3	3	2	3	3	2	3																																																																														
25MCA303.5	-	3	2	2	2	3	3	3	1	2																																																																														
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TEXT BOOKS:																																																																																								
1.	Seema Acharya and Subhashini C: Big Data and Analytics, 2 nd Edition, Wiley India Pvt. Ltd, 2015																																																																																							
2.	Cloud Computing: Principles and paradigms By Raj Kumar Buyya, James Broberg, Andrezei M. Goscinski, 2013																																																																																							
REFERENCE BOOKS:																																																																																								
1.	Judith Hurwitz, Alan Nugent, Fern Halper, Marcia Kaufman: Big data for dummies																																																																																							
2.	Tom White: Hadoop – The Definitive Guide																																																																																							
3.	Chuck Lam: Hadoop in action																																																																																							
4.	Cloud Computing: A Practical Approach by J.Vette, Toby J. Vette, Robert Elsenpeter (Tata McGraw Hill)																																																																																							
5.	Virtualization Complete: Business Basic Edition" by Anil Desai (First Edition)																																																																																							
E Books / MOOCs/ NPTEL																																																																																								
1.	Big Data computing-nptel																																																																																							
2.	Cloud Computing and Big Data, C. Catlett, W. Gentzsch, L. Grandinetti, IOS Press, 2013																																																																																							

REACT NATIVE AND KOTLIN FOR MOBILE APPLICATION DEVELOPMENT			
Course Code	25MCA311	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Assess the difference between native, hybrid and cross-platform application development		
2.	Develop applications based on user controls and apply the debugging and testing functionalities		
3.	Understand the fundamentals of Kotlin and Understand the variables, operators, control flow, and lambda.		
4.	Understand and Analyze Object Oriented capabilities and the UI components in Kotlin		
5.	Understand the concept of internal storage, external storage and SQLite concepts.		
UNIT-I			
Introduction			04 Hours
Different types of Mobile app development platforms: Native, Hybrid, Web applications, Introduction to React Native, React Native: Information flow, Architecture, Threading model. Setting up an environment for developing iOS and Android apps, Introduction to JSX, First Application with create-react-app and Expo, React lifecycle methods, The folder structure			
Basic Components			04 Hours
React Native Basic components: View, Layouts, Touch events, Accessibility, Text, StatusBar, Images, and media. Basic user interaction: Button, List View, Handling user input, TextInput, Additional controls			
UNIT-II			
Debugging and Testing React Native:			05 Hours
Debugging your React Native apps, Remote debugging, Logging, Inspecting React Native components. Testing: Introduction to the Jest testing framework, Snapshot testing your React Native components, working with functions, mocking modules			
Platform APIs:			03 Hours
Using Geolocation, Accessing the User's Images and Camera, Storing Persistent Data with AsyncStorage. Modules and Native Code.			
UNIT-III			
Android Introduction			02 Hours
Setting up Kotlin Environment (Android). Environment Familiarization: - User Interfaces, AVD, Emulator, Testing, Android Studio Code Editor. Android Architecture.			
Kotlin			06 Hours
Kotlin: - Kotlin & Java, Java to Kotlin, Kotlin & Android Studio, Semi-Colon in Kotlin. Kotlin Datatypes, Kotlin Variables, Variable Declaration, Type Annotation & Inference, Nullable, Safe Call Operator, Not Null Assertion, Nullable Type and let Function, Elvis Operator, Type Casting & Checking. Operators & Expression: - Expression, Assignment, Arithmetic, Augmented Assignment, Increment & Decrement, Equality, Boolean Logical, Range & Bitwise. Flow Control (Loops & condition). Functions & Lambda: - Type, Return Values, Parameters, Variable Number, Lambda, Higher Order.			

Basics Of OOP's, Inheritance & Sub classing, Activity Life Cycles, Activity State Changes, Save & Restore State.											
UNIT-IV											
Core Components and UI Handling										08 Hours	
Views, View Groups & Layouts, Android Event Handling, Multi Touch Event Handling, Fragments, Menus & Overflow Menus. Activities, Intents, Broadcast (Intents & Receivers), Content Providers, Manifest, Resource, Service - Thread & Async Task, Started & Bound Service.											
UNIT-V											
Storage										05 Hours	
Shared Preferences, Internal and External storage, Introduction to SQLite Database, CRUD operations using SQLite Database.											
Applications of android classes										03 Hours	
Introductions to Notifications, Using sound vibration and lights, Video Recording & Image Capture, Media Player.											
Course Outcomes: At the end of the course student will be able to											
1.	Differentiate between native, hybrid, and cross-platform mobile application development approaches based on performance, development complexity, and platform compatibility.										
2.	Design and implement user-interactive mobile applications using Kotlin and React Native, incorporating debugging and testing techniques to ensure performance and reliability.										
3.	Demonstrate understanding of Kotlin programming fundamentals including variables, data types, operators, control structures, functions, and lambda expressions.										
4.	Apply object-oriented programming principles in Kotlin to build modular mobile applications and implement UI components using Android's XML and programmatic layouts.										
5.	Implement mobile data persistence techniques using internal storage, external storage, Shared Preferences, and SQLite databases in Kotlin-based Android applications.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA311.1		3	2	-	-	-	-	-	-	3	-
25MCA311.2		-	3	3	-	2	-	-	-	2	3
25MCA311.3		3	3	-	2	-	-	-	-	3	-
25MCA311.4		-	3	3	-	3	-	-	-	2	3
25MCA311.5		-	2	-	2	3	-	2	-	-	3
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	React Native - Building Mobile Apps with JavaScript, Vladimir Novick, PACKT Publishing Limited, 2017										
2.	Building Native Mobile Apps with JavaScript, Bonnie Eisenman, Learning React Native, 2nd Edition, O'Reilly Media Inc., 2017										
3.	Kotlin for Android Developers: Learn Kotlin while developing an Android App, Antonio Leiva , CreateSpace Independent Publishing, 2016										
4.	Android Development with Kotlin, Marcin Moskala, Igor Wojda , Packt Publishing										

REFERENCE BOOKS:	
1.	Mastering React Native, Eric Masiello, Jacob Friedmann, Packt Publishing, 2017
2.	Programming Kotlin, Stephen Samuel, Stefan Bocutiu, Packt Publishing Limited, 2017
3.	Mastering Android Development with Kotlin, Milos Vasic, Packt Publishing Limited, 2017

ANGULAR AND SPRING BOOT			
Course Code:	25MCA312	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisites :	Student must have basic knowledge of HTML, Java Script, Java		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand Single-page application architecture, the Building blocks of Angular and Components used in Angular		
2.	Understand Services and Dependency Injection in Angular, RxJS Observables, Routing and Navigation in Angular		
3.	Understand how to create Forms in Angular and HTTP-client Service in Angular		
4.	Understand Spring Core/DI-IOC, Spring MVC and Spring Boot		
5.	Understand creation of REST services in Spring, Spring Data JPA and creation of Microservices		
UNIT - I			
Single-page application architecture, Angular Building Blocks, Components			02 Hours
Single-page application architecture			
Single-page applications, Single-page application architecture, Introduction to TypeScript, Introduction to Angular, Building your first Angular app.			
Angular Building Blocks			03 Hours
Component architecture, Angular components, Understanding binding, Understanding directives, Pipes			
Components			03 Hours
Exploring components, Lifecycle hooks, Inter-component communication, View encapsulation, Change detection			
UNIT - II			
Services and Dependency Injection, RxJS Observables, Routing and Navigation			02 Hours
Services and Dependency Injection			
Introduction to services, Angular services in depth, Dependency injection			
RxJS Observables			03 Hours
Synchronous vs. reactive (asynchronous) programming, Event handlers, Observables, RxJS operators			
Routing and Navigation			03 Hours
Revisit single page application, Routing terminology, Sample application: Book-master/detail-demo, Route-params, Route order, Adding a redirect			
UNIT - III			
Forms in Angular, HTTP-client Service, Angular Modules and Standalone Components			03 Hours
Forms in Angular			
Overview of forms, Key differences, Data flow in forms, Form validations, Creating dynamic forms,			
HTTP-client Service			04 Hours
Overview of server communication, Requesting data from server, Configuring URL parameters, Sending data to the server, Handling request errors, Intercepting requests and responses			
Angular Modules and Standalone Components			02 Hours
Introduction to stand-alone components, Lazy loading, Bootstrapping with standalone component, Configuring dependency injection, Signals., Introduction to NgRx			

UNIT - IV											
Spring Core/DI-IOC, Spring MVC, Spring Boot										04 Hours	
Spring Core/DI-IOC											
Spring framework overview, DI/IOC container, XML configuration and annotation-based configurations, BeanFactory and ApplicationContext											
Spring MVC										02 Hours	
Introduction to Spring MVC, Understanding MVC architecture, Front controller design pattern, DispatcherServlet, Spring (Web) MVC architecture, Spring web components											
Spring Boot										02 Hours	
Java configuration, Understanding spring boot, Auto-configuration, Related annotations, Starters, Embedded containers, Externalized configuration											
UNIT - V											
Spring REST, Spring Data JPA, Microservices										02 Hours	
Spring REST											
Web services and REST introduction, REST architecture, Spring REST and related annotations, Spring REST, @RestController, Actions, Forwarding, Redirection											
Spring Data JPA										03 Hours	
Object relational mapping, Java Persistence API, JPARepository, @Repository, Configuration in application.properties											
Microservices										02 Hours	
Introduction to Microservices, Creating a Set of Cooperating Microservices, Deploying Our Microservices Using Docker, Adding Persistence, Developing Reactive Microservices											
Course Outcomes: At the end of the course student will be able to											
1.	Implement Single-page application architecture, the Building blocks of Angular and Components used in Angular										
2.	Implement Services and Dependency Injection in Angular, RxJS Observables, Routing and Navigation in Angular										
3.	Implement how to create Forms in Angular and HTTP-client Service in Angular										
4.	Implement Spring Core/DI-IOC, Spring MVC and Spring Boot										
5.	Implement creation of REST services in Spring, Spring Data JPA and creation of Microservices										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓Course Outcomes									1	2
	25MCA312.1	3	2	2	3	3	3	2	3	3	2
	25MCA312.2	3	2	2	3	3	3	2	3	3	2
	25MCA312.3	3	2	2	3	3	3	2	3	3	2
	25MCA312.4	3	2	2	3	3	3	2	3	3	2
	25MCA312.5	3	2	2	3	3	3	2	3	3	2
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Full Stack Development with Angular and Spring Boot: Build scalable, responsive, and dynamic enterprise-level web applications Paperback – 18 July 2024 by Sangeeta Joshi										
2.	Hands-On Microservices with Spring Boot and Spring Cloud Paperback – 20 September 2019 by Magnus Larsson										

REFERENCE BOOKS:

1.	Learning Angular, by Aristeidis Bampakos
2.	Mastering Angular for Web Development: Build Dynamic, Scalable Web Apps with Angular Framework by Rafael Sanders
3.	Spring Start Here: Learn what you need and learn it well by Laurentiu Spilca
4.	Spring in Action, Sixth Edition 6th ed. Edition by Craig Walls
5.	Spring Boot: Up and Running: Building Cloud Native Java and Kotlin Applications 1st Edition by Mark Heckler
6.	Microservices with Spring Boot and Spring Cloud - Second Edition: Build resilient and scalable microservices using Spring Cloud, Istio, and Kubernetes 2nd ed. Edition by Magnus Larsson

UI/UX DESIGN & PRODUCT MANAGEMENT			
Course Code:	25MCA313	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To adopt User centered design thinking and understand the User research methods.		
2.	Apply design thinking to empathise with users, define problem statements and ideate a product using wireframes.		
3.	Conduct Usability studies and learn fundamentals of Visual Design		
4.	Build a high fidelity prototype of your product, conduct user research, Test and iterate the design.		
5.	Understand the fundamentals of Product management.		
UNIT - I			
Fundamentals of UX Design:			04 Hours
Difference between UI and UX, Principles of Good Design, Design Sprints, Key Elements of UX. The UX Design Process: Stages of UX design Process: Research, Design, Prototyping, Testing, Implementation.			
UX research:			04 Hours
Types of User Research: Qualitative & Quantitative Integrating research into the design process, Choosing the right research method, Bias in UX research, Areas of Enquiry, Translating research into actionable insights.			
UNIT - II			
			04 Hours
Personas and User Journeys: Creating a Persona, Empathy Maps, Identifying user pain points, Using Personas in the Design Process, Creating User Stories and Scenarios, Problem statements, Hypothesis statements, Design ideation, Translating Requirements into Design Specifications Competitive audits. Interaction Design: Principles of Interaction Design, Designing User Interactions and Flows, Techniques for Interaction Design			
Build wireframes and Low Fidelity prototypes:			04 Hours
Difference Between Wireframes and Prototypes, User flow, Storyboarding, Paper wireframes, Digital wireframing using Figma, Avoiding deceptive patterns.			
UNIT - III			
			04 Hours
Visual Design: Key Principles of Visual Design, Typography, Color Theory, Images, Creating a Cohesive Visual Language. Accessibility in UX Design: Importance of Accessibility in UX Design, WCAG (Web Content Accessibility Guidelines), WCAG Levels of Compliance, Americans with Disabilities Act (ADA), Tools and Techniques for Designing Accessible Experiences.			
			04 Hours

Usability Testing: Types of Usability Testing, Planning and Conducting Usability Tests.
A/B Testing: Principles of A/B Testing, Designing and Running A/B Tests, Using A/B Test Results to Improve UX

UNIT - IV

High fidelity Prototype : **04 Hours**

Gestures and motion, Accessibility considerations, Build a high-fidelity prototype using Figma.

Maintaining and Evolving UX: **04 Hours**

Key Principles of Continuous UX Improvement, Gathering User Feedback Post-Launch, Iterating on Design Based on User Feedback and Analytics

UNIT - V

Product Management: **05 Hours**

Understanding Product, Product Manager Role and day-to-day Responsibilities, Product Development Lifecycle, Product Vision & Strategy Foundations, Product Requirements Document (PRD), Lean Product Development, Agile, Scrum, Kanban, Competitive and Market Analysis, Roadmapping, Minimum Viable Product.

Product Analytics: **03 Hours**

Acquisition metrics, Activation Metrics, Engagement Metrics, Retention Metrics, Revenue Metrics, Metrics Framework: HEART Metrics Framework, AARRR (Pirate) Metrics Framework

Course Outcomes: At the end of the course student will be able to

1.	Apply user centered design thinking to understand users through User Research.
2.	Use design thinking to empathise with users, define problem statements and ideate a product using wireframes.
3.	Conduct Usability studies and understand fundamentals of Visual Design
4.	Design a high fidelity prototype of a product by conducting user research.
5.	Understand how to build products, evaluate & implement them.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA313.1	1	3	3	3	3	3	3	3	3	3
25MCA313.2	1	3	3	3	3	3	3	3	3	3
25MCA313.3	2	3	3	3	3	3	3	3	3	3
25MCA313.4	2	3	3	3	3	3	3	3	3	3
25MCA313.5	1	3	3	3	3	3	3	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. **The Ultimate UX Designer's Handbook: Master User-Centered Design, Use AI to Your Advantage, Build a Portfolio, Balancing Multiple Roles and Responsibilities ... to Excel in Your Role - By Valentina Stenhamn, Mixon Stenham, 2024**

2.	The Product Book -How to become a great Product Manager, by Product School, Josh Anon and Carlos Gonzalez De Villaumbrosia, 2nd Edition
REFERENCE BOOKS:	
1.	The Design of Everyday Things - Don Norman,by ABC Creations
2.	About Face: The Essentials of Interaction Design by Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel
3.	Elements of User Experience, by Jesse James Garrett, Published by New Riders
4.	Lean UX: Designing Great Products with Agile Teams 2nd Edition,by Jeff Gothelf & Josh Seiden
E Books / MOOCs/ NPTEL	
1.	Google UX Design Professional Certificate, Google Career Certificates, Coursera
2.	Become a Product manager, Course on UDEMY, Instructor: Cole Mercer, Evan Kimbrell
3.	IBM Product Manager Professional Certificate, Coursera by John Rofrano, Daniel C. Yeomans, Matt Versdahl and Rav Ahuja.

HEALTH CARE ANALYTICS

Course Code	25MCA321	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Mathematical Foundation for computer Applications Data Warehousing and Data Mining		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Understand the role that data analytics plays in healthcare operations.
2.	Be familiar with the characteristics of healthcare data- associated data systems- and analytics tools.
3.	Understand the organizational needs and process to enact healthcare data analytics
4.	Understand the Biomedical Information Mining and clinical applications
5.	Mining of statistical model and visual analytics for health care.

UNIT-I

Fundamentals of Healthcare Analytics

06 Hours

Components of healthcare analytics, Data and information, Measurement, Metrics and Indicators. From data to knowledge. Analytics building blocks. Descriptive, predictive, and prescriptive analytics. Healthcare analytics applications, an introduction.

Medical Images and Analytics

04 Hours

Overview of Digital Imaging and Communications in Medicine (DICOM) standards. Biomedical image analysis, biomedical Imaging modalities, object detection, image segmentation, image registration- feature extraction.

UNIT-II

Mining of sensor data in healthcare

08 Hours

Challenges of healthcare data analysis. Biomedical signal analysis, types of biomedical signals, correlation analysis.

UNIT-III

Genomic Data and Analytics

07 Hours

Genomic data analysis for personalized medicine, genomic data generation, Microarray data, Next Generation Sequencing, Public Database, Predictive models for integrating clinical and genomic data, introduction, issues and challenges in integrating clinical and genomic data, different types of integration.

UNIT - IV

Biomedical Information Mining and Social Media Analytics

06 Hours

Electronic Health Records (EHR), components of EHR, coding systems, benefits and challenges of EHR. HL7(Health Level 7). Natural language processing, core NLP Components, Morphological Analysis, Lexical Analysis, Syntactic Analysis, Semantic Analysis, Data Encoding, mining information from clinical text- challenges of processing clinical reports, clinical applications.

Social media analysis for healthcare

03 Hours

Social media analysis for detection and tracking of infectious disease. Social Media Analysis for Public Health Research.

UNIT - V

Advanced Healthcare Data Analytics

04 Hours

Basic statistical prediction models, Alternative clinical prediction models, Survival models, Evaluation and validation. Temporal data mining for healthcare data, Association analysis, Temporal pattern mining.

Visual analytics for healthcare										02 Hours	
Visual analytics for clinical workflow- visual analytics for clinicians and patients.											
Course Outcomes: At the end of the course student will be able to											
1.	Explain the fundamental concepts of healthcare analytics, including data measurement, analytics methodologies, and applications in healthcare decision-making										
2.	Analyze biomedical images and sensor-based healthcare data using signal processing, object detection, and feature extraction techniques										
3.	Evaluate genomic data and its integration with clinical information for predictive modeling and personalized medicine										
4.	Apply data mining, NLP, and social media analytics to extract insights from healthcare records and public health data.										
5.	Develop and validate healthcare predictive models using statistical techniques, temporal data mining, and visual analytics for clinical decision support.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA321.1	-	3	2	1	1	-	2	1	3	2
	25MCA321.2	-	3	2	1	1	-	2	1	3	2
	25MCA321.3	-	3	2	1	1	-	2	1	3	2
	25MCA321.4	-	3	2	1	1	-	2	1	3	2
	25MCA321.5	-	3	2	1	1	-	2	1	3	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	C.K. Reddy and C.C. Aggarwal, Healthcare Data Analytics 2015.										
2.	Christo ElMorr and Hossam Ali-Hassan, Analytics in Healthcare A Practical Introduction, 2019.										
REFERENCE BOOKS:											
1.	Adam Bohr and Kaveh Memarzadeh, “Artificial Intelligence in Healthcare”, Academic Press, 2020.										
2.	H. Yang and E.K. Lee, Health care Analytics: From Data to Knowledge to Healthcare Improvement, 2016.										
3.	Richard Bibb, Dominic Egg beer and Abby Paterson, Medical Modelling, “The Application of Advanced Design and Rapid Prototyping Techniques in Medicine”, 2 nd Edition, 2015.										
E Books / MOOCs/ NPTEL											
1.	E-Book: Healthcare Analytics Made Simple.										
2.	BIG DATA ANALYTICS FOR HEALTHCARE-source: Source: www.coursera.org										
3.	http://www.hl7.org/										
4.	HL7 Messaging Standard Version 2.8.2 - http://www.hl7.org/implement/standards/product_brief.cfm?product_id=185										

GENERATIVE AI AND PROMPT ENGINEERING

Course Code	25MCA322	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Basic understanding of machine learning and programming in Python.		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand the evolution and principles behind Generative AI models.
2.	To implement deep learning-based generative models using TensorFlow.
3.	To apply LSTM and Transformer-based models for text and media generation.
4.	To explore prompt engineering techniques for interacting with LLMs.
5.	To design, tune and evaluate effective prompts using ethical practices.

UNIT-I

Fundamentals of Generative AI: **08 Hours**

Overview of Generative AI: discriminative vs generative models, Historical evolution: RBMs, VAEs, GANs, Mathematical foundations: probability, latent variables, Use cases: image synthesis, music, chatbots, TensorFlow setup, GPU usage, reproducibility

UNIT-II

Image Generation with VAEs and GANs : **08 Hours**

Variational Autoencoders (VAEs) and implementation, Generative Adversarial Networks (GANs): architecture and training, StyleGAN, pix2pix, deepfakes, Image-to-image translation and transfer

UNIT-III

Text, Music & Transformer-based Generation : **08 Hours**

Sequence models (RNN, LSTM) for text/music generation, Transformers and Attention mechanism, Fine-tuning pre-trained models (BERT, GPT-2), Creative AI: MuseGAN, game-play AI, hybrid architectures

UNIT - IV

Prompt Engineering Fundamentals : **08 Hours**

Introduction to Prompt Engineering, Types: zero-shot, one-shot, few-shot prompts, In-context learning and CoT (Chain-of-Thought) prompting, Role-based prompts and structure templates

UNIT - V

Advanced Prompting and Evaluation : **08 Hours**

Prompt tuning and optimization, RAG (Retrieval-Augmented Generation) and prompt chaining, Handling hallucination and ambiguity, Evaluation metrics (BLEU, ROUGE), ethics and safety

Course Outcomes: At the end of the course student will be able to

1.	Understand the foundational concepts of Generative AI and model categories.
2.	Implement VAEs and GANs for image generation using TensorFlow.
3.	Apply sequence models and transformers for text and media generation.
4.	Design and apply effective prompt engineering techniques.
5.	Evaluate, tune, and ethically manage prompt-based GenAI systems.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	
↓ Course Outcomes									1	2	
25MCA322.1	3	2	--	--	--	--	--	--	3	--	
25MCA322.2	3	3	2	2	--	--	--	--	3	2	
25MCA322.3	2	3	3	3	2	--	--	--	3	3	
25MCA322.4	2	2	3	3	3	--	2	--	2	3	
25MCA322.5	3	2	3	3	3	--	2	2	3	3	
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Joseph Babcock, Raghav Bali, 'Generative AI with Python and TensorFlow 2', Packt Publishing, 2023.										
2.	Dennis Rothman, 'Prompt Engineering for Generative AI', O'Reilly Media, 2023.										
REFERENCE BOOKS:											
1.	Anirudh Koul, Siddha Ganju, Meher Kasam, 'Generative AI with LangChain, LLMs, and ChatGPT', Packt, 2024.										
2.	Tom Taulli, 'Generative AI: Navigating the Future of Artificial Intelligence', Wiley, 2023.										
3.	OpenAI Documentation – https://platform.openai.com/docs										
4.	Hugging Face Documentation – https://huggingface.co/docs										

NATURAL LANGUAGE PROCESSING			
Course Code	25MCA323	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To introduce the fundamentals of NLP and provide an understanding of text preprocessing techniques such as tokenization, stemming, lemmatization, and handling noisy text.		
2.	To familiarize students with word representations and similarity techniques, including n-grams, TF-IDF, and word embeddings for semantic analysis.		
3.	To develop expertise in probabilistic language models, including Hidden Markov Models (HMM), Part-of-Speech (POS) tagging, and Named Entity Recognition (NER).		
4.	To explore syntactic parsing and machine translation, covering various parsing techniques and state-of-the-art transformer-based models such as BERT and GPT.		
5.	To apply NLP techniques in real-world applications, including information extraction, sentiment analysis, event detection, and model deployment.		
UNIT - I			
Introduction to Natural Language Processing:			04 Hours
History, Scope, and Challenges of NLP, Applications of NLP in Real-World Scenarios, NLP Pipeline and its Phases, Cross-lingual and Multilingual NLP			
Text Processing & Morphological Analysis:			04 Hours
Regular Expressions for Text Processing, Tokenization (Word, Sentence, Subword), Text Normalization (Lemmatization, Stemming), Handling Noisy Text (Stopword Removal, Lowercasing), Processing Indian Languages in NLP			
UNIT - II			
Spelling Correction & Text Similarity:			04 Hours
Minimum Edit Distance, N-Grams – Counting Words in Corpora, Simple (Un-smoothed) N-Grams, Applications of Spelling Correction (Autocomplete, Search)			
Word Representation & Embeddings:			04 Hours
Word Meaning Representation (Vector Semantics), Cosine Similarity for Measuring Word Similarity, TF-IDF: Weighing Terms in Documents, Introduction to Word Embeddings (Word2Vec, fastText)			
UNIT - III			
Language Models & Sequence Labelling:			04 Hours
Applications of Language Modeling (Speech Recognition, Chatbots), Named Entity Recognition (NER)			
Part of Speech (POS) Tagging:			04 Hours
English Word Classes & Tag Sets, Hidden Markov Models (HMMs) for POS Tagging, Viterbi Algorithm for Sequence Prediction			

UNIT - IV											
Syntactic Parsing										05 Hours	
Grammars and Syntax Structure, Parsing Strategies: Top-Down vs. Bottom-Up Tree Parsing, Depth-First vs. Breadth-First Parsing, Chart Parsers & Transition Networks											
Machine Transformer Models:										03 Hours	
Pre-trained Transformer Models (BERT, GPT, T5)											
UNIT - V											
NLP Applications:										04 Hours	
Information Extraction, Relation Detection and Classification, Sentiment Analysis, Text Summarization											
Temporal & Event Processing:										04 Hours	
Temporal Information Extraction, Event Detection and Classification, Template Filling for Event Processing											
Course Outcomes: At the end of the course, student will be able to											
1.	Understand the fundamentals of Natural Language Processing (NLP) and basic text preprocessing techniques.										
2.	Learn to perform finite-state morphological parsing, detect and correct spelling errors, implement n-gram models and understand Hidden Markov Models (HMM).										
3.	Implement text similarity and word representation techniques for NLP applications.										
4.	Develop language models and perform Part-of-Speech (POS) tagging using probabilistic methods										
5.	Apply NLP techniques for information extraction, event processing, and sentiment analysis, and understand their real-world applications										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA323.1	3	3	3	3	-	-	3	3	3	3
	25MCA323.2	3	3	3	3	-	-	3	3	3	3
	25MCA323.3	3	3	3	3	-	-	3	3	3	3
	25MCA323.4	3	3	3	3	-	-	3	3	3	3
	25MCA33.5	3	3	3	3	-	-	3	3	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Jurafsky, D. and J. H. Martin, ”Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, 2 nd Edition, Prentice Hall, 2008										
2.	Allen, James, “Natural Language Understanding”, 2 nd Edition, Benjamin/Cumming, 1995										
REFERENCE BOOKS:											
1.	Steven Bird, S., Klein, E., Loper, E, “Natural Language Processing with Python-Analyzing Text with the Natural Language Toolkit”, O'Reilly Media, 2010										
2.	Grant S Ingersoll, Thomas S. Morton, and Andrew L. Farris,” Taming text: how to find, organize, and manipulate it” Manning Publications Co., 2013										

3.	Feldman Ronen, and James Sanger,” The text mining handbook: advanced approaches in analyzing unstructured data”, Cambridge university press,2007
4.	Christopher D Manning, and Hinrich Schütze,” Foundations of statistical natural language processing”, MIT press, 1999
E Books / MOOCs/ NPTEL	
1.	http://tamingtext.com/2013/01/11/taming-text-print-copies-now-available/
2.	Youtube Lecture for Indian Language Processing: Natural Language Processing for Indic Languages – SciPy India 2020 https://web.stanford.edu/~jurafsky/slp3/

INDUSTRIAL AND MEDICAL IoT			
Course Code	25MCA331	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Data Communication and Networks		
Teaching Department : Master of Computer Applications			
Course Objectives:			
1.	Understanding the fundamental characteristics, physical and logical design, communication models, and relevant protocols of the Internet of Things		
2.	Understanding of the principles, architectures, and enabling technologies of IIoT and Industry 4.0, enabling them to analyze and apply these concepts in industrial settings		
3.	Students will understand how medical IoT and WBAN technologies can be used in healthcare		
4.	To equip students with the foundational knowledge and practical skills to understand, design, and program IoT endpoints using sensors, actuators, Raspberry Pi, and Arduino		
5.	Students will gain a broad understanding of IoT applications in homes, cities, industries, and healthcare		
UNIT - I			
Introduction to IoT:			04 Hours
Characteristics of IoT			
Physical Design of IoT:			
Things in IoT, IoT protocol			
Logical Design of IoT:			
Functional blocks of IoT, Communication models & APIs			
Developing IoTs:			04 Hours
IoT design methodology, Case study on IoT system for home automation, Case study on IoT system for weather monitoring			
UNIT - II			
Introduction and Architecture of IIoT & Industry 4.0			04 Hours
IIoT and Key Differences: IoT vs IIoT, Prerequisites of IIoT			
Industry 4.0: Design requirements, Drivers, Smart business perspective			
Architecture of IIoT: Categorization of reference architecture in IIoT, Industrial Internet Reference Architecture (IIRA) basics, framework			
Technologies of IIoT:			04 Hours
Off-Site Technologies: Necessity of Cloud Computing, Cloud Computing and IIoT, Cloud Computing Services in IIoT			
Fog Computing for IIoT: Applications and solutions			
On-Site Technologies:			
Augmented Reality: History, Categorization, Applications			
Virtual Reality: History, Categorization, Application			
Big Data and Advanced Analytics: Acquisition, Storage, Necessity of Data Analytics			

UNIT - III											
Introduction to Medical IoT										03 Hours	
Healthcare 4.0 & Smart Hospitals, MIIoT Devices: Wearables, Smart Implants, Remote Patient Monitoring											
Role of IoT in Wireless Body Area Networks (WBAN):										06 Hours	
Introduction to WBAN, Hardware architecture of biomedical sensors, Methods for deployment of biomedical sensors, Data dissemination of sensory information, Requirements of BMSs in WBAN, Applications of Medical Sensors, Data Dissemination Protocols in WBAN, Routing protocols, Medium access control protocols, Data Dissemination Architecture using Healthcare-IoT, Open Issues in WBAN											
UNIT - IV											
IoT Physical Device & Endpoint :										03 Hours	
Sensors, actuators, and smart objects: Definition and role of sensors in IoT, Classification of sensors, Characteristics of sensor, Types of sensors. Actuators: Definition and role of actuators in IoT. Smart Objects: Definition and characteristics of smart objects.											
Basic Building Blocks of an IoT Device										06 Hours	
Introduction to Raspberry Pi: Board details, Overview of Raspberry Pi models, Programming Raspberry Pi with Python: Python basics for IoT, GPIO programming using the RPi.GPIO, Interfacing sensors and actuators with Raspberry Pi using Python., Reading sensor data and controlling actuators, Working with external libraries											
Introduction to Arduino Board: Programming Arduino devices, C/C++ basics for Arduino, Arduino programming structure (setup(), loop()),Digital and analog I/O programming, Interfacing sensors and actuators with Arduino, Using Arduino libraries for sensors and actuators.											
UNIT - V											
Applications of IoT, IIoT and MIIoT :										06 Hours	
Applications of IoT: Case Study on Home Automation, Case Study on Smart Parking, Case Study on Smart Irrigation.											
Applications of IIoT: Smart Manufacturing, Smart Factory: Characteristics, Technologies used in Smart Factories											
Applications of MIIoT: Remote Patient Monitoring, Smart Hospitals											
Course Outcomes: At the end of the course student will be able to											
1.	Articulate the core concepts of IoT, including its characteristics, physical and logical designs, and the role of various communication protocols.										
2.	Able to analyze and apply the principles, architectures, and technologies of IIoT and Industry 4.0 to solve industrial challenges.										
3.	Understand how to use medical IoT and WBAN to create healthcare solutions, and they will be able to discuss the benefits and challenges of these technologies.										
4.	Learn to connect sensors and actuators, program Raspberry Pi and Arduino, and understand how IoT devices work										
5.	IoT is used in real-world situations like smart homes, hospitals and industries and will be able to understand and design these systems.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA331.1	3	1	1	1	1	1	1	1	2	3
	25MCA331.2	2	2	1	2	1	1	1	1	2	3
	25MCA331.3	2	2	1	2	1	1	1	1	2	3

	25MCA331.4	3	3	2	3	3	2	1	2	2	3
	25MCA331.5	3	2	3	2	2	1	2	2	2	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Vijay Madiseti, Arshdeep Bahga, “Internet of Things: A Hands -On Approach”										
2.	Valentina Emilia Balas et al., Internet of Things in Healthcare and Wellbeing, 2020										
3.	Introduction to Industrial Internet of Things and Industry 4.0" by Sudip Misra, Chandana Roy, and Anandarup Mukherjee, 2021										
REFERENCE BOOKS:											
1.	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes										
2.	IoT in Healthcare Systems: Applications, Benefits, Challenges, and Case Studies, 2024, Piyush Kumar Shukla, Aditya Patel, Prashant Kumar Shukla , Prashant Parashar										
3.	Hugh Boyes*, Bil Hallaq, Joe Cunningham, Tim Watson “The industrial internet of things (IIoT): An analysis framework”. Published by Elsevier										
E Books / MOOCs/ NPTEL											
1.	https://www.splunk.com/en_us/data-insider/what-is-the-internet-of-medical-things-iomt.html#iomt-and-telehealth										

.NET FRAMEWORK AND C#			
Course Code	25MCA332	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the .NET framework.		
2.	Understand the concept of class hierarchy and method overloading. Overriding abstract classes and interfaces exception handling I/O stream classes.		
3.	Understand the use of events and delegates Use of multi-threading.		
4.	Understanding multithreading and I/O stream concepts.		
5.	Data access with ADO.NET		
UNIT-I			
Introduction to the .NET Platform			03 Hours
Benefits of .NET Framework Architecture of .NET Framework 4.0 Components of .NET Framework 4.0: CLR CTS Metadata and Assemblies .NET Framework Class Library Windows Forms			
Introducing C#			04 Hours
Need of C# C# Pre-processor Directives Creating a Simple C# Console Application Identifiers and Keywords. Data Types Variables and Constants: Value Types Reference Types Type Conversions Boxing and Unboxing Variables and Constants. Operator Precedence Using the ?? (Null Coalescing)Operator Using the :: (Scope Resolution) Operator and Using the is and as Operators			
UNIT-II			
Classes Objects and Structures			02 Hours
Classes and Objects: Creating a Class Creating an Object Using this Keyword Creating an Array of Objects Using the Nested Classes Defining Partial Classes and methods Static Classes and Static Class Members Structures: Syntax of a struct and Access Modifiers for structs Enumerations.			
Arrays and Strings			01 Hours
Arrays Multidimensional arrays jagged arrays Assigning array references Foreach loop Strings of operator overloading			
Methods			01 Hours
Controlling access to class members Pass objects to methods Using ref and out parameters Using a variable number of arguments Returning objects Method overloading Constructors Garbage collection and destructors Overloading constructors.			
Properties Object- Oriented Programming Concepts			03 Hours
Read-only Property Static Property Accessibility of accessors and Anonymous types. Indexers Encapsulation: Encapsulation using accessors and mutators Encapsulation using Properties. Extension methods. Polymorphism: Compile time Polymorphism/ Overloading Runtime Polymorphism/ Overriding. Abstraction: Abstract classes Abstract methods. Interfaces: Syntax of Interfaces Implementation of Interfaces Interfaces and Inheritance			

UNIT-III	
Delegates Events and Exception Handling	05 Hours
Delegates: Creating and using Delegates Multicasting with Delegates. Events: Event Sources Event Handlers Events and Delegates Multiple Event Handlers. Exception Handling: System exception class Exception handling fundamentals Consequences of an uncaught exception Using multiple catch statements Catching all exceptions Nested try blocks Throwing an exception Using finally Commonly used exceptions.	
Namespaces the Preprocessor and Assemblies	03 Hours
Namespaces Preprocessor Assemblies and the internal access modifier; Runtime Type ID Reflection and Attributes: Runtime type definition Reflection Using Reflection Attributes Using the built-in attributes.	
UNIT - IV	
Multithreaded Programming	03 Hours
Multithreading Fundamentals Thread class Determining the end of Thread IsBackground property Thread priorities Synchronization Thread communication using Wait() Pulse() PulseAll() UsingMethodImplAttribute Suspending resuming stopping threads Determining threads state.	
Working with Collections, Building Components	02 Hours
Collections overview, Collection interfaces, Dictionary entry structure, General-purpose collection class, Storing bits and BitArray, Specialized collections, Accessing a collection using an enumerator. Introduction to components, Overview of C# component, IComponent, Simple component, Overriding Dispose(), Employing the using statement, Containers.	
Using I/O	02 Hours
C#'s I/O built in streams Stream classes Console I/O FileStream and Byte-Oriented file I/O Character based file I/O Redirecting the standard streams Reading and writing binary data Using memory stream Using StreamReader and StreamWriter.	
Unit-V	
Graphical User Interface with Windows Forms	04 Hours
Handling Mechanism Another Way to Create Event Handlers Locating Event Information.Introduction Windows Forms Event Handling: A Simple Event- Driven GUI Visual Studio Generated GUI Code Delegates and Event- Control Properties and Layout Labels TextBoxes and Buttons GroupBoxes and Panels CheckBoxes and RadioButtons ToolTips Mouse-Event Handling Keyboard-Event Handling	
Database Access with ADO.NET	04 Hours
Data Access with ADO.NET: The need for ADO.NET Faces of ADO.NET Role of ADO.NET Data Providers Understanding the ADO.NET Namespaces Types of System.Data Examining the DataColumn type Examining the DataRow Type Details of the DataTable Building a complete DataTable Understanding the DataView type Understanding the role of the DataSet Expressing relations using the DataRelation type Building a simple Test Database Selecting a Data Provider Types of the System.Data.OleDb Namespace Working with the Connected Layer of ADO.NET Working with the OleDbDataReader Inserting updating and deleting records using OleDbCommand Executing a stored Procedure using OleDbCommand Disconnected layer and the OleDbDataAdapter type Working with the SQL Data Provider.	

Web Development Using .Net										03 Hours																																																																														
Web Applications Using MVC.: Understanding the MVC Pattern Creating Controllers and Actions Routing Mechanisms.																																																																																								
Creating Controllers and Actions: Guides on defining controllers and their action methods to handle HTTP requests.																																																																																								
Routing Mechanisms: Details how routing works in ASP.NET Core MVC to map URLs to controller actions.																																																																																								
Course Outcomes: At the end of the course student will be able to																																																																																								
1.	To understand .net framework and C# program structure																																																																																							
2.	To perform polymorphism and to use abstract classes																																																																																							
3.	Use memory stream string reader and string writer, delegates and events																																																																																							
4.	To create multiple threads communication																																																																																							
5.	To understand .net framework and C# program structure																																																																																							
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes																																																																																								
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TEXTBOOKS:																																																																																								
1.	Herbert Schildt, The Complete Reference C#, TATA McGraw-Hill edition																																																																																							
2.	Andrew Troelsen, C# and .NET Platform, Apress,																																																																																							
REFERENCE BOOKS:																																																																																								
1.	Ben Albahari, Peter Drayton and Brad Merril, "C# Essentials".																																																																																							
2.	"Microsoft C# Language Specifications", WP Publishers and Distributors Pvt. Ltd.																																																																																							
E Books / MOOCs/ NPTEL																																																																																								
1.	The little ASP.NET core book by Natte barbettini.																																																																																							
2.	C# 7.0 in a Nutshell: The Definitive Reference																																																																																							

DIGITAL AND SOCIAL MEDIA MARKETING			
Course Code:	25MCA333	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the concept of Digital marketing and Content Marketing.		
2.	Describe critical components of Social Media Marketing and digital advertising.		
3.	Understand how to market your product using email communications and social media influencers.		
4.	Apply Search Engine optimization techniques to improve the digital presence.		
5.	Apply key metrics in Digital Marketing for tracking and measuring success		
UNIT - I			
Introduction:			04 Hours
Understanding Digital Marketing, Traditional Marketing V/s Digital Marketing. Digital Marketing Funnel, Digital Marketing Techniques,Digital Marketing Channels,Digital Marketing Objectives, Digital Marketing Strategies.			
Content Marketing Framework			04 Hours
Techniques of Well-Developed Content Marketing, Full Marketing Funnel for content marketing, Create Intent-Based Content, Ascension Focused Content Marketing, Segmented Content Marketing, Complete Content Marketing is Cross Channeled,Smart Content Development Tool, Content Marketing Plan, Tactics to Measure Content Marketing Success			
UNIT - II			
Digital Advertising			04 Hours
Paid Traffic, Best Sources for Paid Traffic, How to Implement Digital Advertising Easily, Concepts of Evaluating Your Target Audience, Five Elements of High Performing Advertisement Campaigns, How to Implement and Scale Your Ad Campaign, Metrics to Measure Your Digital Advertising Campaign			
Critical Components of Social Media Marketing:			04 Hours
Social Listening, Social Influencing, Social Networking, Social Selling, Social Media Success Metrics			
UNIT - III			
E-mail Marketing			04 Hours
Roles of E-mail Marketing, Methods of Well-Executed E-mail Marketing, Types of Effective Marketing E-mails, The Phases of E-mail Marketing, The Metrics to Measure E-mail Marketing			
The art of influencer marketing			04 Hours
Why use influencers for marketing, Influencer types, The rise of the virtual influencer, Selecting social media influencers,Collaborating with social media influencers.			
UNIT - IV			
SEO strategy for a digital presence			08 Hours

The importance of search engines, Working with search engine algorithms, The ethics of search engine optimisation, Competitor and market research, Ethical and sustainability considerations, How search engines work, Implementing search engine optimisation, Keyword clustering and cannibalisation, Measuring SEO strategy efforts.

UNIT - V

Key Metrics in Digital Marketing for Tracking and Measuring Success

04 Hours

Why Data Is Important, Categories of Metrics in Digital Marketing, Measuring Your Digital Marketing Metrics Successfully, Digital Marketing Metrics You Need to Track and Measure, Traffic Metrics, Conversion Metrics, Revenue Metrics

Conversion Rate Optimization (CRO)

04 Hours

Conversion Rate Optimization (CRO), Why Is CRO So Important? ,How to Perfectly Execute CRO, The Metrics

Course Outcomes: At the end of the course student will be able to

1.	Describe the concept of Digital marketing and Content Marketing
2.	Identify critical components of Social Media Marketing and digital advertising.
3.	Summarize different social media and marketing communication methods.
4.	Apply Search Engine optimization techniques to improve the page rankings.
5.	Analyze digital marketing using different metrics for tracking and measuring success

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA333.1	-	3	3	3	-	3	-	3	3	3
25MCA333.2	-	3	3	3	-	3	-	3	3	3
25MCA333.3	-	3	3	3	-	3	-	3	3	3
25MCA333.4	3	3	3	3	-	3	-	3	3	3
25MCA333.5	3	3	3	3	-	3	-	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. “Digital Marketing Essentials: You always wanted to know”, Vibrant Publishers, 2020
2. Heinze, Aleksej; Fletcher, Gordon; Cruz, Ana; Fenton, Alex (2024-08-12T22:58:59.000). Digital and Social Media Marketing: A Results-Driven Approach (p. xiii). Taylor & Francis 3rd Edition

REFERENCE BOOKS:

1. Andrew T. Stephen, Yakov Bart, “Social Media Marketing- Principles and Strategies”, Stukent, 2015
2. Alex Fenton, Mostafa Mohamad, Ashley Jones “Digital and Social Media Marketing: A Results-Driven Approach”, 2nd Edition, Rottledge.

E Books / MOOCs/ NPTEL

1. Digital Marketing - By Shainesh G | Indian Institute of Management Bangalore (IIMB), NPTEL, 2025

FULL STACK WEB DEVELOPMENT USING MERN LAB										
Course Code			25MCA304			Course Type			PCC	
Teaching Hours/Week (L: T: P: S)			0:0:4:0			Credits			02	
Total Teaching Hours			26			CIE + SEE Marks			50+50	
Teaching Department: Master of Computer Applications										
Course Objectives:										
1.	Understand the need for server side rendered apps.									
2.	Use a JavaScript package manager (either npm or Yarn).									
3.	Making HTTP requests to read or modify data.									
4.	Students can understand how callbacks and MongoDB function.									
5.	Students able to create CRUD operations, routing mechanisms, and UI components.									
List of Experiments										
1.	Design a resume of a job aspirant using React components like Classes and Functions. Style the resume by applying CSS.									
2.	Build Student Registration Portal using entities like component, State and Props									
3.	Design an employee Management system using RESTFULL APIs in React.									
4.	Deploy connectivity between React and Node Application for Inventory Management system.									
5.	Demonstrate Node Js Application to perform CRUD operation for online Book Cart.									
6.	Development of an application using RESTFULL API's									
Course Outcomes: At the end of this course, the students will be able to:										
1.	Develop dynamic and interactive user interfaces using React by leveraging JSX transformation, state management, event handling, and efficient UI component integration for modern web applications.									
2.	Understand and implement full-stack web development by applying the concepts of MVC architecture, utilizing Node.js, Express, and MongoDB for backend development, and building RESTful APIs with CRUD operations									
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes										
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes									1	2
25MCA304.1	3	3	3	3	3	2	1	1	3	3
25MCA304.2	2	3	3	3	3	1	1	1	3	2
1: Low 2: Medium 3: High										
REFERENCE BOOKS:										
1.	Vasan Subramanian – Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, 2 nd Edition, Apress, 2019.									
2.	Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, Tyler McGinnis – Full Stack React: The Complete Guide to ReactJS and Friends, Fullstack.io, 2017.									
3.	Alex Banks, Eve Porcello – Learning React: Modern Patterns for Developing React Apps, 2 nd Edition, O'Reilly Media, 2020.									
4.	Mario Casciaro, Luciano Mammino – Node.js Design Patterns: Master Best Practices to Build Modular and Scalable Server-Side Web Applications, 3 rd Edition, Packt Publishing, 2020.									
5.	Shivakumar Babji – Full Stack Web Development: HTML, CSS, JavaScript, Bootstrap, and PHP, Independently Published, 2023.									
6.	Alok Kumar – Practical Full Stack Machine Learning. BPB Publications. 2021.									

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB											
Course Code				25MCA305			Course Type			PCC	
Teaching Hours/Week (L: T: P: S)				0:0:4:0			Credits			02	
Total Teaching Hours				26			CIE + SEE Marks			50+50	
Prerequisite				Foundation for computer Applications							
Teaching Department: Master of Computer Applications											
Course Objectives:											
1.	To understand and implement data preprocessing techniques.										
2.	To understand and implement Supervised and Unsupervised Machine Learning Algorithms.										
3.	To understand and implement Deep Learning Algorithms.										
4.	To understand and implement performance evaluation techniques for the model built.										
List of Experiments											
1.	Write basic programs (Conditional, branching, looping, Methods and modules, classes and objects, numpy, pandas, sklearn) using Python										
2.	Write Programs to demonstrate supervised and unsupervised learning										
3.	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.										
4.	Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.										
5.	Write a program to implement the Naive Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.										
6.	Assuming a set of documents that need to be classified, use the Naive Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.										
7.	Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Java/Python ML library classes/API.										
8.	Apply k-Means algorithm to cluster a set of data stored in a .CSV file. You can add Java/Python ML library classes/API in the program.										
9.	Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.										
10.	Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs										
Course Outcomes: At the end of the course student will be able to											
1.	Apply data preprocessing techniques, supervised and unsupervised machine learning algorithms.										
2.	Apply deep learning algorithms and implement performance evaluation techniques.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA305.1		3	3	3	3	3	-	3	-	3	3
25MCA305.2		3	3	3	3	3	-	3	-	3	3
1: Low 2: Medium 3: High											

REFERENCE BOOKS:	
1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, “ Machine Learning”, 1 st Edition, McGraw Hill Education, 2017.
3.	Elaine Rich, Kevin K and S B Nair, “Artificial Intelligence”, 3 rd Edition, McGraw Hill Education, 2017
4.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
5.	Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2 nd Edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016